

CO3-AT-1-Case Study Questions

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1) Case Study 1 — Employee Salaries

A company records the annual salaries (in ₹ lakhs) of 9 employees: 4, 5, 5, 6, 7, 8, 8, 9, 30

1. Find the mean, median, and mode.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Identify the outlier.
5. Which measure of central tendency best represents the typical salary? Explain why

Solution : Given Data

Employee	Salary (₹ Lakhs)
1	4
2	5
3	5
4	6
5	7
6	8
7	8
8	9
9	30

Number of Employees (n) = 9

1. Mean, Median and Mode

A) Mean

Formula

$$\text{Mean} = \frac{\sum x}{n}$$

Calculation

Total salary:

- $4 + 5 = 9$
- $9 + 5 = 14$
- $14 + 6 = 20$
- $20 + 7 = 27$
- $27 + 8 = 35$
- $35 + 8 = 43$
- $43 + 9 = 52$
- $52 + 30 = 82$

Therefore,

$$\sum x = 82$$

$$\begin{aligned}\text{Mean} &= \frac{82}{9} \\ &= 9.1\end{aligned}$$

Mean = ₹9.11 lakhs

Interpretation:

The average salary is ₹9.11 lakhs. However, the salary of ₹30 lakhs increases the average, making it higher than the salaries earned by most employees.

B) Median

Arrange the salaries in ascending order.

4, 5, 5, 6, 7, 8, 8, 9, 30

Since

$$n = 9$$

Median position

$$\begin{aligned} &= \frac{9 + 1}{2} \\ &= \frac{10}{2} \\ &= 5^{th} \text{ observation} \end{aligned}$$

The 5th value is 7.

Median = ₹7 lakhs

Interpretation:

The median indicates that half the employees earn below ₹7 lakhs and half earn above it. It is not affected by the outlier.

C) Mode

Salary	Frequency
4	1
5	2
6	1
7	1
8	2
9	1
30	1

The highest frequency is 2.

Therefore,

Mode = ₹5 lakhs and ₹8 lakhs

The dataset is bimodal because two values occur most frequently.

2. Range

Formula

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Highest salary = 30

Lowest salary = 4

$$\begin{aligned}\text{Range} &= 30 - 4 \\ &= 26\end{aligned}$$

Range = ₹26 lakhs

Interpretation:

The salary difference between the highest-paid and lowest-paid employee is ₹26 lakhs, indicating a wide variation in salaries.

3. Variance and Standard Deviation

Mean = 9.11

Salary (x)	$x - 9.11$	$(x - 9.11)^2$
4	-5.11	26.11
5	-4.11	16.89
5	-4.11	16.89
6	-3.11	9.67
7	-2.11	4.45
8	-1.11	1.23
8	-1.11	1.23
9	-0.11	0.01
30	20.89	436.39

Variance

$$\sum (x - \bar{x})^2 = 512.87$$

$$\begin{aligned}\text{Variance} &= \frac{512.87}{9} \\ &= 56.99\end{aligned}$$

Variance = 56.99

Standard Deviation

$$= \sqrt{56.99}$$

$$= 7.55$$

Standard Deviation = ₹7.55 lakhs

Interpretation:

The high standard deviation indicates that the salaries are widely spread due to the unusually high salary of ₹30 lakhs.

4. Identification of the Outlier

An outlier is a value that is significantly different from the rest of the data.

Most employee salaries range between ₹4 lakhs and ₹9 lakhs, whereas one employee earns ₹30 lakhs.

Outlier = ₹30 lakhs

Interpretation:

The salary of ₹30 lakhs is much higher than the remaining salaries. It increases the mean, variance, and standard deviation, resulting in a positively skewed distribution.

5. Best Measure of Central Tendency

Measure Value

Mean ₹9.11 lakhs

Median ₹7 lakhs

Mode ₹5 lakhs and ₹8 lakhs

Among these measures, the median (₹7 lakhs) is the most appropriate measure of central tendency because it is not affected by the outlier. The mean is inflated by the salary of ₹30 lakhs, whereas the median represents the salary of the middle employee and gives a more realistic picture of the typical salary.

2) Case Study 2 — Website Response Time

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Calculate the mean and median.

2. Find the range.

3. Calculate the variance and standard deviation.

4. Determine whether the data is likely to be right-skewed or left-skewed.

5. Explain why the median may be more useful than the mean.

Solution: Given Data

Website response times (in milliseconds):

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

Number of observations (n) = 10

1. Mean and Median

A) Mean

Formula

$$\text{Mean} = \frac{\sum x}{n}$$

Calculation

Total response time:

- $120 + 125 = 245$
- $245 + 128 = 373$
- $373 + 130 = 503$
- $503 + 132 = 635$
- $635 + 135 = 770$
- $770 + 140 = 910$
- $910 + 145 = 1055$
- $1055 + 150 = 1205$
- $1205 + 300 = 1505$

Therefore,

$$\begin{aligned}\sum x &= 1505 \\ \text{Mean} &= \frac{1505}{10} \\ &= 150.5 \text{ ms}\end{aligned}$$

Mean = 150.5 ms

Interpretation:

The average website response time is 150.5 milliseconds. However, the unusually high response time of 300 ms increases the average.

B) Median

Since there are 10 observations (even number),

$$\text{Median} = \frac{n^{th} + \left(\frac{n}{2} + 1\right)^{th}}{2}$$

$$= \frac{5^{th} + 6^{th}}{2}$$

The ordered data is:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

Median

$$\begin{aligned} &= \frac{132 + 135}{2} \\ &= \frac{267}{2} \\ &= 133.5 \text{ ms} \end{aligned}$$

Median = 133.5 ms

Interpretation:

The median response time is 133.5 ms, which better represents the typical website response because it is not affected by the extreme value.

2. Range

Formula

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Maximum response time = 300 ms

Minimum response time = 120 ms

$$\begin{aligned} \text{Range} &= 300 - 120 \\ &= 180 \text{ ms} \end{aligned}$$

Range = 180 ms

Interpretation:

The response times vary by 180 milliseconds, indicating a large difference between the fastest and slowest request.

3. Variance and Standard Deviation

Mean = 150.5 ms

Response Time (x)	$x - 150.5$	$(x - 150.5)^2$
120	-30.5	930.25
125	-25.5	650.25
128	-22.5	506.25
130	-20.5	420.25
132	-18.5	342.25
135	-15.5	240.25
140	-10.5	110.25
145	-5.5	30.25
150	-0.5	0.25
300	149.5	22350.25

$$\sum (x - \bar{x})^2 = 25581.50$$

Variance

$$\begin{aligned}\text{Variance} &= \frac{25581.50}{10} \\ &= 2558.15\end{aligned}$$

Variance = 2558.15

Standard Deviation

$$\begin{aligned}&= \sqrt{2558.15} \\ &= 50.58 \text{ ms}\end{aligned}$$

Standard Deviation = 50.58 ms

Interpretation:

The high standard deviation indicates that the response times vary considerably because of the unusually slow response time of 300 ms.

4. Skewness of the Data

Most response times lie between 120 ms and 150 ms, while one request has a response time of 300 ms.

Since the extreme value is on the higher side, the distribution has a long tail to the right.

Therefore, the data is positively skewed (right-skewed).

Interpretation:

The high response time of 300 ms pulls the mean towards the right, making the distribution asymmetric.

5. Why is the Median More Useful than the Mean?

The median (133.5 ms) is more useful because it is not affected by the unusually high response time of 300 ms. The mean increases to 150.5 ms due to this outlier, even though most response times are between 120 ms and 150 ms.

Therefore, the median provides a more accurate representation of the website's typical response time and is a better measure of central tendency for this dataset.

3)Case Study 3 — Student Performance

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

- 1. Find Q1, Q2 and Q3.**
- 2. Calculate the IQR.**
- 3. Use the $1.5 \times \text{IQR}$ rule to identify any outliers.**
- 4. Find the mean and median.**
- 5. Explain how the outlier affects the mean.**

Solution:

Given Data

Marks of 12 students:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Number of students (n) = 12

1. Calculation of Q1, Q2 and Q3

Arrange the data in ascending order (already arranged):

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Step 1: Find Q2 (Median)

Since n = 12 (even),

Median is the average of the 6th and 7th observations.

6th observation = 52

7th observation = 55

$$\begin{aligned}Q_2 &= \frac{52 + 55}{2} \\&= \frac{107}{2} \\&= 53.5\end{aligned}$$

Q2 = 53.5

Step 2: Find Q1

The lower half of the data is:

42, 45, 48, 50, 52, 52

Since there are 6 values, Q1 is the average of the 3rd and 4th observations.

$$\begin{aligned}Q_1 &= \frac{48 + 50}{2} \\&= \frac{98}{2} \\&= 49\end{aligned}$$

Q1 = 49

Step 3: Find Q3

The upper half of the data is:

55, 58, 60, 62, 65, 95

Q3 is the average of the 3rd and 4th observations.

$$\begin{aligned}Q_3 &= \frac{60 + 62}{2} \\&= \frac{122}{2} \\&= 61\end{aligned}$$

$$Q_3 = 61$$

2. Calculation of Interquartile Range (IQR)

Formula

$$IQR = Q_3 - Q_1$$

Substitute the values:

$$= 61 - 49$$

$$= 12$$

$$IQR = 12$$

Interpretation:

The middle 50% of the students scored within a range of 12 marks.

3. Outlier Detection Using the $1.5 \times IQR$ Rule

Step 1: Calculate $1.5 \times IQR$

$$1.5 \times 12 = 18$$

Step 2: Find the Lower and Upper Limits

Lower Limit

$$Q_1 - 1.5(IQR)$$

$$49 - 18 = 31$$

Upper Limit

$$Q_3 + 1.5(IQR)$$

$$61 + 18 = 79$$

Therefore,

- Lower Limit = 31
- Upper Limit = 79

Any value less than 31 or greater than 79 is considered an outlier.

Since $95 > 79$,

Outlier = 95

Interpretation:

The score of 95 is much higher than the remaining marks and is identified as an outlier.

4. Mean and Median

Mean 38

Median 38

0

25

50

75

100

No outlier: mean is 38; median is 38

No outlier High outlier

No outlier High outlier

A) Mean

Formula

$$\text{Mean} = \frac{\sum x}{n}$$

Calculation

Total marks:

- $42 + 45 = 87$
- $87 + 48 = 135$
- $135 + 50 = 185$
- $185 + 52 = 237$
- $237 + 52 = 289$
- $289 + 55 = 344$
- $344 + 58 = 402$
- $402 + 60 = 462$
- $462 + 62 = 524$
- $524 + 65 = 589$
- $589 + 95 = 684$

Therefore,

$$\begin{aligned}\sum x &= 684 \\ \text{Mean} &= \frac{684}{12} \\ &= 57\end{aligned}$$

Mean = 57

B) Median

From the earlier calculation,

Median (Q2) = 53.5

5. Effect of the Outlier on the Mean

The score 95 is much higher than the marks of the other students.

Because of this high score:

- The mean increases to 57.
- The median remains 53.5, as it depends only on the middle values.
- Therefore, the mean is influenced by the outlier, whereas the median is not.

Hence, if the objective is to represent the typical student performance, the median is more reliable than the mean for this dataset.

4)Case Study 4 — E-Commerce Orders

An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

- 1. Calculate the mean, median, and mode.**
- 2. Find the variance and standard deviation.**
- 3. Calculate the coefficient of variation if the mean and standard deviation are known.**
- 4. What does the standard deviation tell the company?**
- 5. Would you describe the order data as highly or moderately variable?**

Solution : Given Data

Number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

Number of observations (n) = 10

1. Mean, Median and Mode

A) Mean

Formula

$$\text{Mean} = \frac{\sum x}{n}$$

Calculation

Total orders:

- $18 + 20 = 38$
- $38 + 22 = 60$
- $60 + 22 = 82$
- $82 + 24 = 106$
- $106 + 25 = 131$
- $131 + 26 = 157$
- $157 + 28 = 185$
- $185 + 30 = 215$
- $215 + 35 = \mathbf{250}$

Therefore,

$$\begin{aligned}\sum x &= 250 \\ \text{Mean} &= \frac{250}{10} \\ &= 25\end{aligned}$$

Mean = 25 orders

B) Median

Since $n = 10$ (even),

Median is the average of the 5th and 6th observations.

Ordered data:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

$$\begin{aligned}\text{Median} &= \frac{24 + 25}{2} \\ &= \frac{49}{2} \\ &= 24.5\end{aligned}$$

Median = 24.5 orders

C) Mode

Orders	Frequency
18	1
20	1
22	2
24	1
25	1
26	1
28	1
30	1
35	1

The highest frequency is 2.

Mode = 22 orders

2. Variance and Standard Deviation

Mean = **25**

Orders (x)	$x - 25$	$(x - 25)^2$
18	-7	49
20	-5	25
22	-3	9
22	-3	9
24	-1	1
25	0	0
26	1	1
28	3	9
30	5	25
35	10	100

Variance

$$\sum (x - \bar{x})^2 = 228$$

$$\text{Variance} = \frac{228}{10} \\ = 22.8$$

Variance = 22.8

Standard Deviation

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

The points have mean $\mu = 5$ and $\sigma = 1.5$.

σ

σ

Drag points

$$= \sqrt{22.8} \\ \approx 4.77$$

Standard Deviation = 4.77 orders

3. Coefficient of Variation (CV)

$CV = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100$

Formula

Substitute the values:

$$CV = \frac{4.77}{25} \times 100 \\ = 19.08\%$$

Coefficient of Variation = 19.08%

Interpretation:

A coefficient of variation of 19.08% indicates that the daily orders show a moderate level of variation compared to the average number of orders.

4. What Does the Standard Deviation Tell the Company?

The standard deviation measures how much the daily order numbers differ from the average.

A standard deviation of 4.77 orders means that the number of daily orders generally varies by about 5 orders from the average of 25 orders. This information helps the company estimate fluctuations in demand and plan inventory, staff, and delivery resources efficiently.

5. Is the Order Data Highly or Moderately Variable?

The Coefficient of Variation (19.08%) is below 20%, indicating that the variation in daily orders is moderate rather than high.

Therefore, the order data is moderately variable, suggesting that daily sales are relatively consistent with only small fluctuations.

5)case Study 5 — Comparing Two Data Science Models

Two machine-learning models have the following prediction errors:

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the mean error for both models.
2. Calculate the variance for both models.
3. Calculate the standard deviation for both.
4. Which model has more consistent predictions?
5. Can the model with the lower mean error necessarily be considered better? Explain.

Solution: Given Data

Model A Errors

2, 3, 3, 4, 4, 5, 5, 6

Model B Errors

1, 1, 2, 4, 5, 7, 8, 9

Number of observations (n) = 8

1. Mean Error for Both Models

Mean38

Median38

0

25

50

75

100

No outlier: mean is 38; median is 38

No outlierHigh outlier

No outlierHigh outlier

Model A

Formula

$$\text{Mean} = \frac{\sum x}{n}$$

Calculation

Total errors:

- $2 + 3 = 5$
- $5 + 3 = 8$
- $8 + 4 = 12$
- $12 + 4 = 16$
- $16 + 5 = 21$
- $21 + 5 = 26$
- $26 + 6 = \mathbf{32}$

$$\text{Mean} = \frac{32}{8} = 4$$

Mean Error (Model A) = 4

Model B

Total errors:

- $1 + 1 = 2$
- $2 + 2 = 4$
- $4 + 4 = 8$
- $8 + 5 = 13$
- $13 + 7 = 20$
- $20 + 8 = 28$
- $28 + 9 = \mathbf{37}$

$$\begin{aligned}\text{Mean} &= \frac{37}{8} \\ &= 4.625\end{aligned}$$

Mean Error (Model B) = 4.63

2. Variance for Both Models

Model A

Mean = 4

Error (x)	$x - 4$	$(x - 4)^2$
2	-2	4
3	-1	1
3	-1	1
4	0	0
4	0	0
5	1	1
5	1	1
6	2	4

$$\sum (x - \bar{x})^2 = 12$$

$$\text{Variance} = \frac{12}{8} = 1.5$$

Variance (Model A) = 1.5

Model B

Mean = **4.625**

Error (x) $x - 4.625$ $(x - 4.625)^2$

1 -3.625 13.14

1	-3.625	13.14
2	-2.625	6.89
4	-0.625	0.39
5	0.375	0.14
7	2.375	5.64
8	3.375	11.39
9	4.375	19.14

$$\sum (x - \bar{x})^2 = 69.87$$

$$\text{Variance} = \frac{69.87}{8}$$

$$= 8.73$$

Variance (Model B) = 8.73

3. Standard Deviation

Model A

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

The points have mean $\mu = 5$ and $\sigma = 1.5$.

Drag points

$$\begin{aligned} SD &= \sqrt{1.5} \\ &= 1.22 \end{aligned}$$

Standard Deviation (Model A) = 1.22

Model B

$$\begin{aligned} SD &= \sqrt{8.73} \\ &= 2.95 \end{aligned}$$

Standard Deviation (Model B) = 2.95

4. Which Model Has More Consistent Predictions?

Model A has a lower variance (1.5) and lower standard deviation (1.22) than Model B.

This means the prediction errors of Model A are closer to the mean, making its predictions more consistent and stable.

Therefore,

Model A has more consistent predictions.

5. Can the Model with the Lower Mean Error Always Be Considered Better?

No.

A lower mean error alone does not guarantee that a model is better. The consistency of predictions is also important.

A model with a low mean error but very high variation may produce unreliable predictions. Therefore, both the mean error and the standard deviation should be considered before selecting the best model.

In this case:

- Model A: Mean = 4, SD = 1.22
- Model B: Mean = 4.63, SD = 2.95

Since Model A has both a lower mean error and a lower standard deviation, it is the better-performing model.